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ORIGINAL PAPER

Racial differences in the incidence of breast cancer subtypes defined by combined histologic grade and hormone receptor status

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Abstract Breast cancer encompasses several distinct clinical entities of very different characteristics and behaviors, a fact which likely contributes to the higher breast cancer mortality in African-Americans (AA) despite the higher incidence in European-Americans (EA). We are interested in how incidence variability in cancer subtypes defined by combined estrogen receptor (ER) and grade contributes to racial mortality disparities. As an initial step, we compared age-specific and age-adjusted incidence rates for each ER/Grade subtype in South Carolina (SC—a southern state) with Ohio (a northern mid-western state), using state registry data for 1996–2004. Each ER/Grade subtype had a distinct incidence pattern and rate, with three striking racial/geographic differences. First, the racial incidence disparity in ER negative (ER−) cancers was mostly within the ER−/G3 subtype, of which AAs had ~65% higher incidence than did EAs; ER−/G2 was much less common, but of significantly higher incidence in AAs. Second, the racial disparity in ER positive (ER+) cancers was in the ER+/lower-grade cancers, with a marked EA excess in both states. Third, AA incidence of the ER+/lower-grade subtypes was ~26% higher in Ohio than in SC. The other subtypes (ER−/G1 and ER+/G3) varied minimally by race and state, and the latter showed a strong association with age. Age adjustment halved the racial difference in mean age at diagnosis to about 2 years younger in AAs, compared to 4 years younger in case comparisons. Use of age-adjusted and age-specific rates of breast cancer subtypes may improve understanding of racial incidence and mortality disparities over time and geography. This approach also may aid in estimating the race-specific incidence rates of triple-negative breast cancer.

Keywords Breast cancer · Epidemiology · Health status disparities · Estrogen receptor · Histologic grade · Non-SEER registry

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Abbreviations

AA	African-American
ACoS	American College of Surgeons
CDC	Centers for Disease Control and Prevention
dk	Not known or unavailable
EA	European-American
ER	Estrogen receptor
ER+	Positive ER expression status
ER+/G1	ER positive and Grade 1

ER+/G1, 2	ER positive and either Grade 1 or Grade 2
ER−	Negative ER expression status
HER-2/neu	Human epidermal growth factor receptor-2 also called HER-2
NAACCR	North American Association of Central Cancer Registries
NPCR	National Program of Cancer Registries
PR	Progesterone receptor
SC	South Carolina
SCCCR	South Carolina Central Cancer Registry
SEER	Surveillance, Epidemiology and End Results (program of the NCI)
T-stage	Tumor stage, based upon size of invasive primary tumor
vs	Versus

Introduction

Breast cancer encompasses several distinct clinical entities of very different characteristics and behaviors. The greater prevalence of aggressive breast cancer in African-American (AA) breast cancer patients [1, 2] likely contributes to the high AA mortality rates, even though incidence of breast cancer is greater in European-American women (EA). Recognizing that the magnitudes of these contrary racial disparities vary substantially by state [3], we propose this is in part due to variability in the incidence of these distinct clinical entities.

Histologic grade is an important standard prognostic factor [4–6], used along with hormone receptor expression status, histologic type, and tumor stage in patient management and treatment decision making [5, 7–9]. Even with the advent of specific molecular markers such as HER-2/neu, high histologic grade (poorly or undifferentiated) remains an important and strong predictor of poor survival [6]. Most population-based analyses of breast cancer incidence focus on either stage, estrogen receptor (ER) status or (secondly) grade, so examining incidence rates of clinically important subgroups defined by combined ER status plus grade provides the opportunity for a more nuanced description of racial disparities. These subtypes are also useful approximations for the clinically relevant breast cancer entities defined by microarray profiles, particularly the basal-like and luminal cancers, and the triple-negative cancers—absent ER, progesterone receptor (PR) and HER-2/neu expression and overwhelmingly high grade (G3). Given that HER-2/neu data are not yet routinely available from cancer registries, the incidence rates of ER-negative cancers and specifically the subtype ER-negative high grade (ER−/G3) may serve as useful indicators of the triple-negative phenotype incidence.

Using South Carolina hospital data, we previously reported that ER−/G3 cancers were more common in AA patients, even among low T-stages and across age groups [10]. Similar findings have been reported in other case populations [11], including the AA excess of triple-negative breast cancer [12]. However, such analyses are based on comparisons of AA and EA breast cancer cases. They are, therefore, subject to bias by the different age distributions of the populations from which the cases arise, and AA populations are skewed toward younger ages compared to EAs (Fig. 1). Moreover, comparisons of case proportions cannot readily identify the additive effects of opposing trends. These problems are eliminated by examining age-specific and age-adjusted rates, which define the population-based nature of the racial incidence disparities.

To our knowledge, incidence rates of subtypes defined by combined ER and grade have not previously been examined. We hypothesized that the greater AA incidence of ER-negative breast cancer is found primarily in the ER−/G3 subset. We further hypothesized that the EA excess of ER+/lower-grade disease accounts for the greater part of the additional breast cancer incidence variation between AAs and EAs. South Carolina (SC) was selected for study because of its large African-American population, our concern for cancer disparities here, and the high quality of the SC Central Cancer Registry (SCCCR). Ohio was chosen as a comparison state, geographically, culturally, and economically different but historically connected with SC as a terminus of the Underground Railroad, with a comparably sized AA population to SC and with a good state cancer repository.

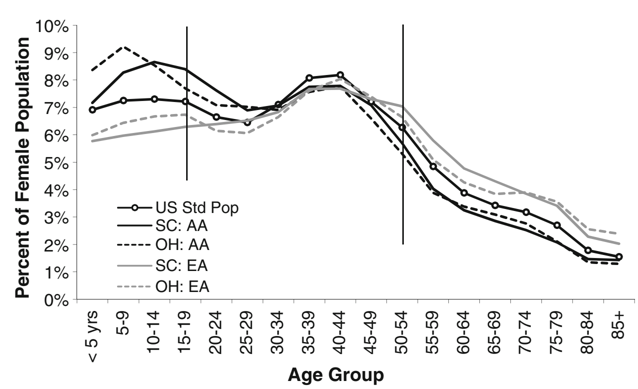

Fig. 1 Age distributions of year 2000 female populations in South Carolina and Ohio, and the 2000 US standard million population. Bars separate age groups 15–49 years and 50–85+ years. AA African-American, EA European-American, SC South Carolina, Std Pop US 2000 Standard Population Sources: State population data: CDC WONDER On-line Database. http://wonder.cdc.gov/bridged-race-v2006.html (16). 2000 US Standard Million Population: Surveillance Research Program, NCI. http://seer.cancer.gov/stdpopulations (18)

Materials and methods

Cancer case datasets

Data were obtained from the South Carolina Central Cancer Registry (SCCCR) and the Ohio Cancer Incidence Surveillance System (OCISS). Although not a SEER registry (Surveillance, Epidemiology and End Results), the SCCCR is Gold Certified by the National Program of Cancer Registries (NPCR), a quality control program administered by the Centers for Disease Control and Prevention (CDC). The latest audit (2005) found the SCCCR to have a completeness (coverage) rate of 97% and an accuracy rate of 96%, both of which exceeded the national standard of 95%. It includes cancers diagnosed in SC residents attending SC facilities and cancers diagnosed in SC residents at institutions in 16 other states, including neighboring states. The SC female population (year 2000) was about 543,000 AAs and 1,235,000 EAs, and a large proportion of each are considered rural. By comparison, the only SEER registry with a large southern rural population at all similar to SC is Rural Georgia, which covers a female population of only about 28,700 AAs and about 32,500 EAs (Census 2000 data).

The Ohio Cancer Incidence Surveillance System (OCISS) is also a participant in the NPCR program. It has Silver Certification, with over 90% complete coverage of cancers in Ohio residents diagnosed and/or treated in Ohio during years 1996–2006.

Individual-level anonymized data for breast cancer (ICD-10 codes C50.0-50.9, ICDO-3 codes 8000-8983) were obtained from the SCCCR for the period 1 January 1996 (when it began collecting cancer data) through 31 December 2004 and from the OCISS for the same time period. Analysis was restricted to AA and EA women because the female population of SC includes only 2% other or unknown races. Hormone receptor expression status and histologic grade were not available for 92% of SC *in situ* cases, so the study population was further restricted to invasive cancers only. As is typical of cancer registries, unavailability of HER-2/neu data precluded specific examination of triple-negative cancers. Disease characteristics and race had been abstracted from medical records by hospital or registry personnel.

Analytic methods

Analyses were conducted using Microsoft Excel and Intercooled Stata 10.1 statistical software (StataCorp LP, College Station, TX). Statistical tests were two-sided with alpha 0.05. Racial and inter-state differences in simple case distributions of demographic and disease characteristics (as in Table 1) were statistically evaluated using chi-square and *t*-tests. Most such comparisons were significant, in part due to large numbers, so comparisons in this context are reported in the text only if $p < 0.001$. Case proportion ratios (e.g., proportion of AA cases that were ER+/G1 compared to the proportion in EAs) were estimated as risk ratios with 95% confidence intervals.

Case number estimates

Tumor stage was available for almost all cases (95% in SC, 93% in Ohio), and Grade for 86% (SC) and 78% (Ohio). Hormone receptor expression data were not required by these non-SEER registries during the study period, so ER status was available for only 47% of SC cases and 59% in Ohio. Data availability was very similar for AAs and EAs within each state.

To accommodate the high degree of data missingness (particularly ER status), we developed a method to impute missing ER and Grade. This allowed us to utilize all cases within each 5-year age stratum, especially important for AAs who only contributed an average of 611 (SC) to 725 (Ohio) cases per year. Age at diagnosis was categorized in 5-year age groups starting at age 15, with ages 85+ combined as one group. Missing ER and/or Grade were imputed on an *aggregate* basis using state-, race-, and stage-specific distributions of known ER and Grade within age group. Stage was categorized as Localized, Regional (extension to lymph nodes only or NOS), Regional (by direct extension), Distant, and Unknown. We assumed that ER and Grade data were missing at random within each such state/race/age/stage stratum. Only 16% of cases were discordant for ER/PR expression, with minimal differences by state or race, so we did not additionally subclassify by PR.

Imputation involved three steps. First, cases with unknown ER status (within each state/race/age/stage stratum) were allocated to ER-positive or ER-negative categories, based on the known ER distribution by Grade. Next, cases with unknown Grade were allocated to well differentiated (G1), moderately differentiated (G2) or poorly/undifferentiated (G3) based upon known Grade distribution by ER status within each stratum. Results of these two steps were added to the stratum-specific tabulations of cases for which these characteristics were both known. Finally, allocation of ER and Grade where both were unknown was achieved based upon this summed ER/Grade distribution (within each stratum).

Incidence rate estimates

State- and race-specific incidence rates were calculated for each ER/Grade subtype. Denominator data were year 2000 state- and race-specific female population estimates (in 5-year age categories) as obtained from the US Census [13]. These estimates of numbers of Whites and Black/African-Americans had used the bridged-race method (Vintage 2006) to collapse the 31 race categories of the 2000 US Census to

Table 1 Disease characteristics of female invasive breast cancer cases in South Carolina and Ohio, 1996–2004

	South Carolina[a]		Ohio[a]	
	AA $n = 5{,}498$ Years (±SD) or n (%)	EA $n = 18{,}420$ Years (±SD) or n (%)	AA $n = 6{,}528$ Years (±SD) or n (%)	EA $n = 64{,}713$ Years (±SD) or n (%)
Age at diagnosis (years)	5,498 (100%)	18,420 (100%)	6,528 (100%)	64,713 (100%)
Mean ± SD	58.1 ± 14.8	62.5 ± 13.9	59.5 ± 14.6	63.3 ± 14.4
Median; range	57; 19–105	63; 19–102	59; 21–104	64; 19–108
SEER summary stage: available	5,187 (94%)[b]	17,583 (95%)[b]	5,952 (91%)[b]	60,334 (93%)[b]
Localized	2,746 (53)	11,630 (66)	3,468 (58)	39,531 (66)
Regional	2,046 (39)	5,264 (30)	2,059 (35)	17,828 (30)
Distant	395 (8)	689 (4)	425 (7)	2,975 (5)
Grade: available	4,673 (85%)	15,755 (86%)	4,938 (76%)	51,187 (79%)
Grade 1: well differentiated	494 (10)	3,375 (21)	673 (14)	11,028 (22)
Grade 2: moderately differentiated	1,466 (31)	6,313 (40)	1,683 (34)	21,497 (42)
Grade 3: poorly differentiated	2,713 (58)	6,067 (39)	2,582 (52)	18,662 (37)
ER status: available	2,651 (48%)	8,609 (47%)	3,935 (60%)	38,033 (59%)
Negative	1,062 (40)	1,985 (23)	1,465 (37)	8,266 (22)
Positive	1,589 (60)	6,684 (77)	2,470 (63)	29,767 (78)
ER status and Grade				
Both ER and Grade available	2,452 (45%)	7,931 (43%)	3,370 (52%)	32,988 (51%)
ER−/G1	20 (<1)	117 (1)	33 (1)	326 (1)
ER−/G2	147 (6)	382 (5)	187 (6)	1,583 (5)
ER−/G3	829 (34)	1,384 (17)	1,066 (32)	5,478 (17)
ER+/G1	244 (10)	1,484 (18)	434 (13)	6,837 (21)
ER+/G2	595 (24)	2,749 (35)	924 (27)	12,296 (37)
ER+/G3	617 (25)	1,815 (23)	726 (22)	6,468 (20)
Either ER or Grade available (not both)	2,420 (44%)	8,502 (46%)	2,133 (33%)	23,244 (36%)
ER−/Gdk	66 (3)	102 (1)	179 (8)	879 (4)
ER+/Gdk	133 (5)	576 (7)	386 (18)	4,166 (18)
ERdk/G1	230 (10)	1,774 (21)	206 (10)	3,865 (17)
ERdk/G2	724 (30)	3,182 (37)	572 (27)	7,618 (33)
ERdk/G3	1,267 (52)	2,868 (34)	790 (37)	6,716 (29)
Neither available (ERdk/Gdk)	626 (11%)	1,987 (11%)	1,025 (16%)	8,481 (13%)
PR status: available	2,635 (48%)	8,508 (46%)	3,895 (60%)	37,517 (58%)
Negative	1,323 (50)	2,995 (35)	1,814 (47)	12,416 (33)
Positive	1,312 (50)	5,513 (65)	2,081 (53)	25,101 (67)
Histologic type (ICD-O-3 code): avail.:	5,498 (100%)	18,420 (100%)	6,528 (100%)	64,713 (100%)
Ductal carcinoma, NOS (8500)	4,012 (73)	13,219 (72)	4,702 (72)	46,064 (71)
Lobular carcinoma, classic type (8520)	265 (5)	1,530 (8)	439 (7)	5,507 (9)
Duct and lobular (8522)	159 (3)	930 (5)	210 (3)	2,697 (4)
Special ductal variants, favorable prognosis[c]	335 (6)	974 (5)	323 (5)	3,193 (5)
Other ductal variants[d]	217 (4)	626 (3)	176 (3)	1,768 (3)
Inflammatory carcinoma (8530)	93 (2)	145 (1)	76 (1)	518 (1)
Other carcinomas[e]	417 (8)	996 (4)	602 (9)	4,966 (8)
Facility type: available	5,498 (100%)	18,420 (100%)	NA	NA
ACoS	3,761 (68%)	12,176 (66%)		
Stage available	3,654 (97)	11,945 (98)		
Grade available	3,336 (89)	10,828 (89)		
ER status available	2,651 (70)	8,609 (71)		

Table 1 continued

	South Carolina[a]		Ohio[a]	
	AA $n = 5{,}498$ Years (±SD) or n (%)	EA $n = 18{,}420$ Years (±SD) or n (%)	AA $n = 6{,}528$ Years (±SD) or n (%)	EA $n = 64{,}713$ Years (±SD) or n (%)
Not ACoS	1,737 (32%)	6,244 (34%)	NA	NA
Stage available	1,533 (88)	5,638 (90)		
Grade available	1,337 (77)	4,927 (79)		
ER status available	0 (0)	0 (0)		

AA African-American, *ACoS* American College of Surgeons, *dk* not known, *EA* European-American, *ER+* estrogen receptor positive, *ER−* estrogen receptor negative, *G* grade, *PR+* progesterone receptor positive, *PR−* progesterone receptor negative, *SD* standard deviation, *SEER* Surveillance, Epidemiology and End Results program of the NCI

[a] All AA vs EA comparisons of tumor characteristic distributions were statistically significant, two-sided $P \leq 0.001$

[b] The first row for each tumor characteristic indicates the number and percent (in parentheses) of all cases for which that characteristic was available. For example, “Grade: available 4,673 (85%)” indicates that grade was available for 4,673 cases (85% of the total 5,498) but missing for 15%. Among this 85%, the numbers and percentages of each grade are given in the following rows and total to 100%. In this example, 494 cases were Grade 1 (11%), 1,466 cases were Grade 2 (31%), and 2,713 cases (58%) were Grade 3

[c] Ductal variants of better prognosis: mucinous, medullary, tubular, papillary, cribriform, adenoid cystic

[d] Other ductal variants: Paget disease and other less common variants including comedocarcinoma, scirrhous, ductular, apocrine, and other ductal variants

[e] Includes carcinomas generally of high grade or poor prognosis, and other minor types

the four categories used by the SC and other registries [14]. Population estimates for each year were unlikely to deviate significantly, and 2000 was the midpoint of the 1996–2004 study period, so this single year estimate was considered adequate. Age-adjusted incidence rates were estimated for each ER/Grade subgroup by the direct method, standardized to the 2000 United States Standard Million Population [15]. Incidence of all subtypes combined was compared to official 2000 state rates available from the North American Association of Central Cancer Registries (NAACCR) [16].

Differences in incidence rates for each ER/Grade subgroup were examined as rate differences (e.g., AA rate minus EA rate) and as rate ratios (AA rate/EA rate). Age-adjusted rate variances were calculated using the method of Armitage and Berry for the Poisson approximation to the binomial variance, and statistical significance of the standardized rate ratios according to the method of Smith [17].

To assess the potential impact of any systematic imputation errors, we conducted a sensitivity analysis, using SC data imputed within state-, race-, and age- (but not stage-) specific strata.

Results

Female population age distributions

In each state, the AA female population age distribution was shifted toward younger ages compared to EAs (Fig. 1). Within the age range of interest, 69% of the female AA populations were younger than age 50, compared to only ~59% of EAs.

Breast cancer case characteristics

Average age at diagnosis was 4 years younger for AAs than EAs in each state ($p < 0.001$; Table 1), consistent with the generally younger age distribution of AA women. Indicators of worse prognosis were more common among AAs; all tumor characteristic distributions were significantly different between AA and EA cases within each state ($p < 0.001$; Table 1). This was true when restricted to cases with known ER and Grade, and also for all cases after imputation. The most aggressive subtype, ER−/G3, comprised about one-third of AA cases but only slightly more than one-sixth of EA cases ($p < 0.0001$) in each state. The less aggressive ER+/G1 and ER+/G2 subtypes together (ER+/G1,2) comprised over half of EA cases. They were substantially more common in EAs than AAs, more so in SC (59%) than Ohio (41%). ER+/G2 was the most common subtype in AA patients of age ≥65 years and in EAs of all ages. At younger ages, ER−/G3 was the predominant subtype in AAs. The other subtypes showed minimal variation in numbers of cases per 100,000 between races or states, but ER−/G2 was about 20% more common in AAs (statistically significant).

In SC, reporting facilities were categorized by the SCCCR as those accredited by the American College of Surgeons (ACoS) and non-ACoS facilities. Generally speaking, ACoS facilities are larger and include a more urban patient catchment. This enabled us to investigate patterns of data missingness and whether they might differentially affect incidence estimates by race based upon type of reporting institution and population served. As shown in Table 1, 68% of AA and 66% of EA cases were reported by ACoS facilities. ER status was only provided by ACoS facilities, and the proportions of cases without stage or Grade, while higher in non-ACoS facilities, were almost identical across race. Diagnosis may have been more timely in ACoS facilities since percent Distant stage was slightly lower than in non-ACoS facilities (4.6% vs 5.2%), but this trend was small and statistically significant only in EAs (3.7 vs. 4.5%, $p = 0.010$). Moreover, our method of imputation incorporated stage. Therefore, we do not expect that missingness by type of reporting institution contributed in any meaningful way to racial differences in subtype incidence rates in SC, unless associations between ER, Grade, and stage varied by facility status. We were not able to evaluate this possibility for Ohio and do acknowledge that this may have limited accuracy of incidence rate estimates in Ohio and comparisons between these two states.

Age-specific and age-adjusted incidence

Distinctly different age-adjusted rates and patterns were found for three of the six ER/Grade combinations (Table 2 and Fig. 2), indicating where within ER-negative and ER-positive groups, the racial differences occur. In each state, AAs had much higher incidence rates of **ER−/G3** compared to EAs ($\sim$65%; $p < 0.001$), representing $\sim$14 excess cases per 100,000 women. This excess was spread over most ages, with a peak at age 55–59 years in SC but distributed more broadly in Ohio (Fig. 3). **ER−/G2** was also significantly more common in AAs but only by about 1.2 cases per 100,000. In striking contrast, EAs had substantially higher rates of both **ER+/G1 and ER+/G2** cancers than did AAs, especially notable in older women. Combined (**ER+/G1,2**) EA incidence was 78% higher than AAs in SC (30 cases/100,000), but only 47% higher in Ohio (23 cases/100,000). Interestingly, the AA rate of ER+/G1,2 cancer was 26% higher in Ohio than SC ($\sim$10 cases/100,000; $p < 0.001$). The other subtypes (**ER−/G1** and **ER+/G3**) did not differ significantly between races. ER+/G3 was $\sim$10% lower in Ohio for both races

Table 2 Breast cancer age-adjusted mean ages at diagnosis and incidence rates by subtype in South Carolina and Ohio, 1996–2004 (2000 US Standard Million Population)

	South Carolina			Ohio			AA	EA
	AA	EA	AA–EA	AA	EA	AA–EA	Ohio−SC	Ohio−SC
Mean age at diagnosis	59.0	61.1	−2.1	60.1	61.4	−1.3	1.1	0.3
ER/Grade	AA	EA	AA/EA Ratio (95% CI)	AA	EA	AA/EA Ratio (95% CI)	Ohio/SC Ratio (95% CI)	Ohio/SC Ratio (95% CI)
ER−/Grade 1	0.8	1.8	0.46 (0.35, 0.60)	1.0	1.2	0.84 (0.65, 1.08)	1.23 (0.82, 1.82)	0.67 (0.58, 0.79)
ER−/Grade 2	7.1	5.8	1.23 (1.08, 1.40)	7.0	5.8	1.20 (1.08, 1.35)	0.99 (0.86, 1.14)	1.01 (0.94, 1.09)
ER−/Grade 3	34.8	21.1	1.65 (1.54, 1.76)	35.1	21.1	1.66 (1.57, 1.76)	1.01 (0.95, 1.08)	1.00 (0.96, 1.04)
ER+/Grade 1	11.1	24.6	0.45 (0.42, 0.49)	15.4	25.2	0.61 (0.58, 0.65)	1.39 (1.12, 1.54)	1.02 (0.98, 1.06)
ER+/Grade 2	27.5	44.3	0.62 (0.59, 0.65)	33.1	46.2	0.72 (0.69, 0.75)	1.20 (1.12, 1.29)	1.04 (1.02, 1.07)
ER+/Grade 3	27.5	28.1	0.98 (0.92, 1.04)	24.9	25.2	0.99 (0.94, 1.05)	0.91 (0.84, 0.98)	0.90 (0.87, 0.93)
ER−/Grade 1,2	7.9	7.6	1.04 (0.93, 1.17)	8.1	7.1	1.14 (1.03, 1.26)	1.02 (1.00, 1.03)	0.93 (0.92, 0.94)
ER+/Grade 1,2	38.6	68.9	0.56 (0.54, 0.58)	48.5	71.4	0.68 (0.66, 0.70)	1.26 (1.19, 1.36)	1.04 (1.01, 1.06)
			EA/AA: 1.78 (1.75, 1.82)			EA/AA: 1.47 (1.46, 1.49)		
ER negative	42.7	28.7	1.49 (1.40, 1.57)	43.2	28.2	1.53 (1.46, 1.61)	1.01 (0.96, 1.07)	0.98 (0.95, 1.02)
ER positive	66.1	96.9	0.68 (0.66, 0.70)	73.5	96.6	0.76 (0.74, 0.78)	1.11 (0.06, 1.16)	1.00 (0.98, 1.02)
All (1996–2004)	108.8	125.7	0.87 (0.84, 0.89)	116.7	124.8	0.94 (0.91, 0.96)	1.07 (1.04, 1.11)	0.99 (0.98, 1.01)
			EA/AA: 1.16 (1.02, 1.31)			EA/AA: 1.07 (1.01, 1.14)		
Official 2000 rates[a]	110.2	130.4	0.85	116.0	128.9	0.90	1.05	0.99

95% CI 95% confidence interval, *AA* African-American, *EA* European-American, *ER+* estrogen receptor positive, *ER−* estrogen receptor negative, *G* grade, *Grade 1,2* Grades 1 and 2 combined

[a] Official state incidence rates are for year 2000, obtained from NAACCR. Source: http://www.cancer-rates.info/naaccr/

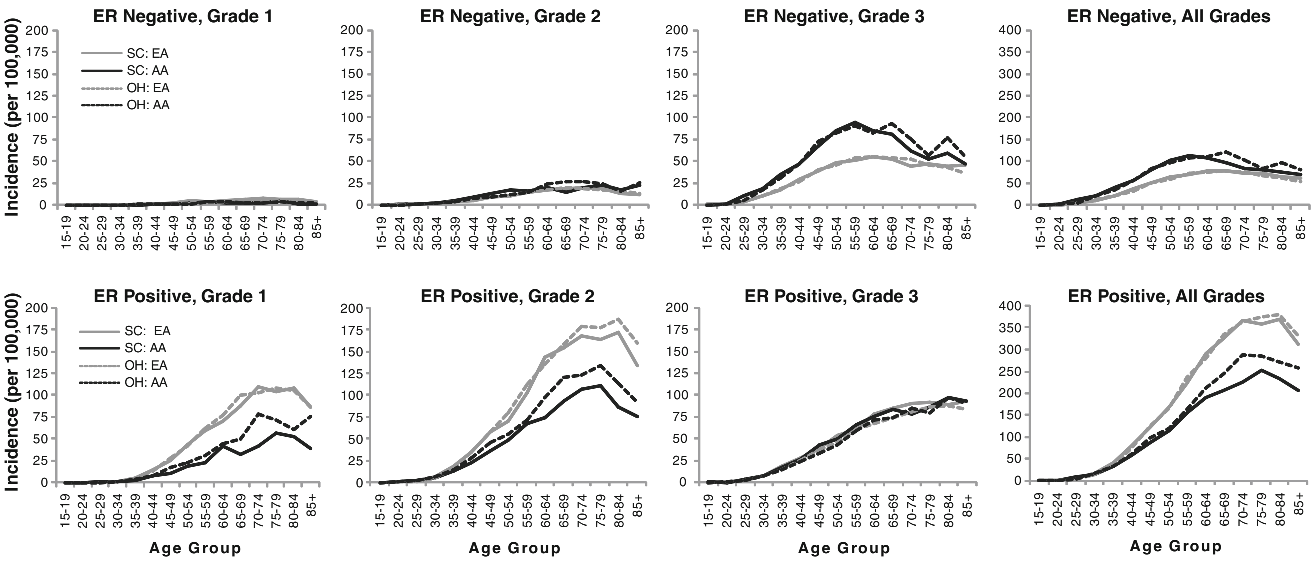

Fig. 2 Age-specific incidence rates (per 100,000) by combined estrogen receptor status and grade. AA African-American, EA European-American, OH Ohio, SC South Carolina

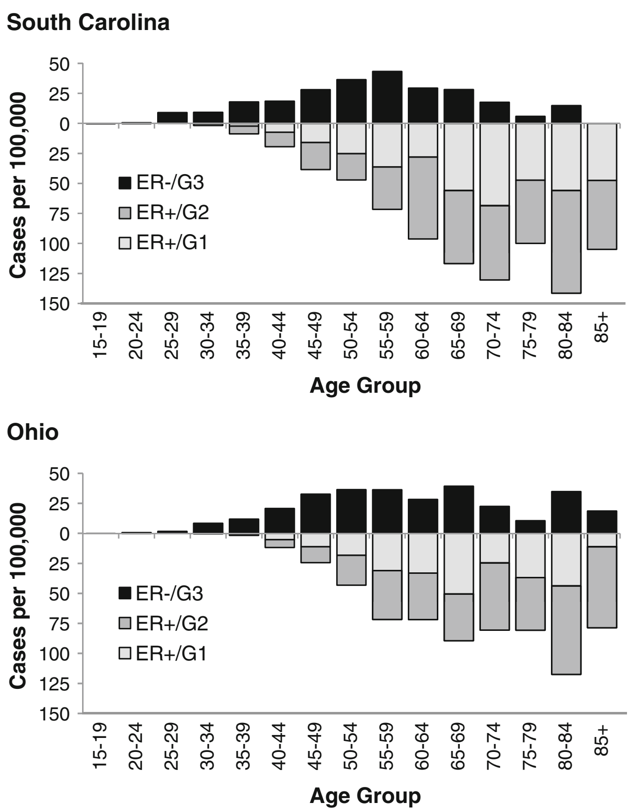

Fig. 3 Age distribution of excess breast cancer incidence (per 100,000) in African-American (AA) compared to European-American (EA) women, by combined estrogen receptor status and grade. Excess incidence was computed by subtracting the incidence in EA women from that in AA women. Bars above the zero line represent excess AA incidence, while bars below the zero line represent excess EA incidence. For clarity, subtypes without statistically significant incidence differences between races have not been plotted (ER−/G1, ER+/G3). ER+ estrogen receptor positive, ER–estrogen receptor negative, G grade

($p < 0.05$ in AAs, $p < 0.001$ in EAs). Rates of **ER−/G1,2** were minimal ($\sim$8 cases/100,000 in each race and state) at every age, although ER−/G2 increased slightly through about age 60–64.

Racial differences in risks of the ER/Grade subtypes were particularly striking when graphically presented as *age-specific rate differences* (AA minus EA). In Fig. 3, bars above the zero line represent the excess incidence in AAs, whereas bars below this line indicate the excess incidence in EAs. In each state, the AAs excess of ER−/G3 cancers is seen at most ages, decreasing after age 55–59 in SC but with a less marked decline in OH. The EA excess incidence of ER+/G1,2 cancers increased with age, especially in SC.

Computed average total age-adjusted incidence rates over the 1996–2004 period (Table 2) were 16% higher in EAs vs AAs in SC (17 cases/100,000), 7% in Ohio (8 cases/100,000). These rates compared very closely to the 2000 official rates.

Our results clearly demonstrate that how the racial difference in mean age at diagnosis and other parameters are quantified makes a difference in subsequent inferences. Specifically, racial difference in mean age at diagnosis was much greater in direct case comparisons than with age adjustment. That is, age adjustment reduced the 4-year racial differences in mean age at diagnosis (no adjustment) to only 1–2 years (Tables 1, 2). Similarly, case proportion ratios overemphasized the excess of ER−/G3 cancer in AAs and underemphasized the ER+/G1,2 excess in EAs (Table 3) when compared with age-adjusted incidence ratios. The magnitude of this effect on inferred racial disparities varied with age, reflecting the differences in AA and EA female population age distributions.

Table 3 Racial differences in case proportions and age-adjusted incidence rates in South Carolina and Ohio 1996–2004, using imputed case numbers

ER/Grade	South Carolina				Ohio			
	Case proportion			Incidence rate	Case proportion			Incidence rate
	AA	EA	AA/EA ratio	AA/EA ratio	AA	EA	AA/EA ratio	AA/EA ratio
All ages								
ER−/Grade 1,2	0.07	0.06	1.21 (1.09, 1.35)	1.04 (0.93, 1.17)	0.07	0.06	1.24 (1.12, 1.36)	1.14 (1.03, 1.26)
ER−/Grade 3	0.33	0.16	2.01 (1.91, 2.11)	1.65 (1.54, 1.76)	0.31	0.16	1.88 (1.81, 1.96)	1.66 (1.57, 1.76)
ER+/Grade 1,2	0.35	0.56	0.63 (0.60, 0.65)	0.56 (0.54, 0.58)	0.41	0.58	0.71 (0.69, 0.73)	0.68 (0.66, 0.70)
ER+/Grade 3	0.25	0.22	1.14 (1.08, 1.20)	0.98 (0.92, 1.04)	0.21	0.20	1.06 (1.01, 1.11)	0.99 (0.94, 1.05)
ER−	0.40	0.22	1.79 (1.72, 1.87)	1.49 (1.40, 1.57)	0.37	0.22	1.72 (1.66, 1.78)	1.53 (1.46, 1.61)
ER+	0.60	0.78	0.77 (0.75, 0.79)	0.68 (0.66, 0.70)	0.63	0.78	0.80 (0.78, 0.82)	0.76 (0.74, 0.78)
Age <50								
ER−/Grade 1,2	0.07	0.06	1.21 (0.98, 1.50)	1.29 (0.94, 1.76)	0.06	0.06	0.98 (0.81, 1.18)	1.06 (0.81, 1.40)
ER−/Grade 3	0.44	0.25	1.73 (1.60, 1.87)	1.82 (1.57, 2.11)	0.42	0.26	1.62 (1.53, 1.72)	1.76 (1.54, 2.00)
ER+/Grade 1,2	0.25	0.43	0.57 (0.52, 0.62)	0.60 (0.53, 0.69)	0.31	0.44	0.71 (0.66, 0.76)	0.78 (0.70, 0.87)
ER+/Grade 3	0.24	0.25	0.96 (0.86, 1.06)	1.01 (0.86, 1.18)	0.20	0.23	0.87 (0.79, 0.95)	0.94 (0.81, 1.09)
ER−	0.51	0.32	1.63 (1.53, 1.74)	1.72 (1.51, 1.96)	0.49	0.32	1.50 (1.42, 1.58)	1.62 (1.44, 1.83)
ER+	0.49	0.68	0.71 (0.67, 0.75)	0.75 (0.68, 0.83)	0.51	0.68	0.76 (0.72, 0.80)	0.84 (0.77, 0.91)
Age ≥50								
ER−/Grade 1,2	0.07	0.06	1.20 (1.06, 1.37)	0.97 (0.59, 1.57)	0.07	0.05	1.33 (1.20, 1.49)	1.17 (0.75, 1.80)
ER−/Grade 3	0.27	0.14	1.95 (1.83, 2.09)	1.55 (1.15, 2.09)	0.26	0.14	1.87 (1.77, 1.97)	1.62 (1.24, 2.11)
ER+/Grade 1,2	0.40	0.58	0.68 (0.65, 0.70)	0.55 (0.47, 0.65)	0.45	0.61	0.74 (0.71, 0.76)	0.66 (0.58, 0.75)
ER+/Grade 3	0.26	0.21	1.20 (1.13, 1.28)	0.97 (0.75, 1.26)	0.22	0.19	1.13 (1.06, 1.19)	1.01 (0.80, 1.27)
ER−	0.35	0.20	1.73 (1.64, 1.83)	1.37 (1.07, 1.77)	0.33	0.19	1.72 (1.64, 1.80)	1.49 (1.19, 1.87)
ER+	0.65	0.80	0.82 (0.80, 0.84)	0.66 (0.58, 0.76)	0.67	0.81	0.83 (0.81, 0.85)	0.74 (0.66, 0.83)

AA African-American, *EA* European-American, *ER+* estrogen receptor positive, *ER−* estrogen receptor negative, *Grade 1,2* Grades 1 and 2 combined

Table 4 Effects of theoretical systematic imputation errors on estimated incidence rates* (using South Carolina data)

	Change in numbers of cases/100,000* (% Change in age-standardized incidence rates)					
	ER−/G1	ER−/G2	ER−/G3	ER+/G1	ER+/G2	ER+/G3
European-Americans						
ER+ overallocated by 10%	−1.2 (−39.4%)	−2.1 (−26.1%)	−1.3 (−5.7%)	1.2 (5.3%)	2.1 (5.1%)	1.3 (4.7%)
G1 overallocated by 10%	0.0 (0.3%)	−0.0 (−0.0%)	0.0 (−0.0%)	0.1 (0.4%)	−0.1 (−0.2%)	−0.0 (−0.2%)
G2 overallocated by 10%	0.0 (−0.1%)	0.0 (0.3%)	−0.0 (−0.1%)	−0.1 (−0.4%)	0.2 (0.4%)	−0.1 (−0.4%)
ER+ overallocated by 20%, and G2 by 30%	−2.5 (−56.6%)	−4.2 (−41.1%)	−2.6 (−11.0%)	2.2 (9.8%)	4.8 (12.3%)	2.2 (8.6%)
African-Americans						
ER+ overallocated by 10%	−0.5 (−36.6%)	−1.3 (−16.7%)	−1.3 (−3.6%)	0.5 (4.7%)	1.3 (5.0%)	1.3 (4.8%)
G1 overallocated by 10%	0.0 (0.3%)	−0.0 (−0.0%)	−0.0 (−0.0%)	0.0 (0.5%)	−0.0 (−0.1%)	−0.0 (−0.1%)
G2 overallocated by 10%	−0.0 (−0.0%)	0.0 (0.3%)	−0.0 (−0.1%)	−0.0 (−0.3%)	0.1 (0.4%)	−0.1 (−0.3%)
ER+ overallocated by 20%, and G2 by 30%	−1.0 (−53.6%)	−2.6 (−28.1%)	−2.6 (−7.2%)	0.9 (8.7%)	3.0 (12.1%)	2.3 (9.0%)

* Change from "true" age-adjusted incidence rate. Note that imputations in this analysis did not include stage, and therefore may exaggerate the potential effects of systematic methodologic errors

To assess the impact of any systematic imputation errors, we conducted a sensitivity analysis using the SC data (Table 4). For computational simplicity, this analysis did not include stage. If our method systematically overallocated cases to ER+ status by 10%, for example, then the incidence rates of each ER+ subgroup would be overestimated by only about 5% (≤1.3 cases/100,000) and ER−/G3 underestimated by ∼4 to 6% (1.3 cases). Of course, rates of ER−/G1 and ER−/G2 subgroups would be substantially underestimated because they are comparatively rare. In contrast, a 10% overallocation error in Grade (the most likely would be erroneous overallocation of G1 and G2 and underallocation of G3) would only cause a <1% corresponding error in incidence (<1 case/100,000 in any ER/G subgroup). A combination of systematic misallocation of ER and grade is also shown. These findings make sense, because about half of SC cases were missing ER status, and only ∼15% were missing histologic grade (ER: 10% misallocation × 50% missing = 5% error; Grade: 10% misallocation × 15% missing = 1.5%, divided between 2 grade categories). We expect that in the Ohio dataset, 10% systematic overallocation to the ER+ category would overestimate the incidence of each ER+ subgroup by ∼4% (10% × 40% missing); 10% overallocation to a Grade would cause a ∼1% error in the other 2 grade categories (10 × 22% missing/2 grade categories). Especially because our imputation method included stratification on stage, we are comfortable that the potential impact of any systematic misimputation resulting from our method is small.

Discussion

This analysis was conducted to clarify breast cancer racial incidence disparities by focusing on rates of prognostically significant subtypes defined by combined estrogen receptor and histologic grade. We specifically focused on disparities in South Carolina and compared them with a geographically and demographically different population with historical connections (Ohio), and demonstrated the advantages of age-specific analysis and age adjustment over case comparisons. Non-SEER registries, such as in South Carolina, provide coverage for important parts of the country that may not be adequately represented in SEER registries for analyses requiring large case numbers from minority populations. Because histologic grade is widely available and prognostically important, NPCR registries may be of practical importance in understanding racial disparities in the occurrence of breast cancer.

We found that of the six ER/Grade subtypes, racial differences existed in only four, particularly in three. The higher AA incidence of ER− cancers was observed primarily in the ER−/G3 subtype and at most ages (less so in ER−/G2). The higher EA incidence of ER+ cancers was only in the ER+/G1 and ER+/G2 subtypes, particularly in older women, and more pronounced in Ohio than in SC. ER−/G1 and G2 cancers were both relatively rare, and ER+/G3 breast cancer increased continuously with age regardless of race or state.

With age adjustment, the typical 4-year racial differences in mean ages at diagnosis were reduced to only 1–2 years. This illustrates the impact of differences in population-at-risk age structure on the age distributions of case groups and serves as a useful reminder to adjust for age when investigating racial disparities. While case proportion comparisons are clinically useful, age-adjusted data are essential when assessing population data. For example, case proportion comparisons underestimated the EA excess of ER+/G1,2 cancers while overestimating the AA excess of ER−/G3, even when cases were separated into age categories (<50 years and ≥50), and could not demonstrate that these racial differences were additive at the population level. Age-specific incidence analysis illustrated that the AA excess of ER−/G3 cancer was distributed over virtually all ages, including older ages (Figs. 2, 3), and was not a phenomenon skewed mostly toward younger women as case proportions would imply.

Breast cancer consists of at least five biologically distinct and clinically relevant entities based on microarray profiles including two that are ER+ (luminal A and B) and three that are ER− (basal-like, HER-2/neu-positive and normal-like [18–20]). The clinical triple-negative and expression-based basal-like phenotypes are not interchangeable [21], but basal-like cancers tend to be histologically high grade and triple negative, with an aggressive clinical behavior and poor prognosis [20]. Similarly, most (80–90%) histologically triple-negative cancers are also high grade [20] and have clinical outcomes similar to basal-like cancers. We found Grade 3 cancers comprised about 83% of all ER-negative cancers in AAs, about 73% in EAs (Table 3). Thus, to the extent that the entire ER-negative group (all grades) or the ER−/G3 subtype specifically demonstrates the triple-negative phenotype, either may serve as a useful indicator of triple-negative incidence in the absence of HER-2/neu registry data.

Recent research supports the concept that ER+ and ER− cancers may originate from different cancer stem cell populations and that tumors of different histologic grade as well as different hormone receptor status may have different biological origins [19, 22–29]. This is corroborated by epidemiologic findings that breast cancer risk-factor profiles vary by tumor hormone receptor status [30–36]. In a recent report, AAs were three times as likely as non-AAs to have triple-negative breast cancer, regardless of age and BMI [36], reminiscent of the typically triple-negative or basal-like phenotypes associated with BRCA1 mutations or inactivation [21, 37] and consistent with a genetic predisposition or other biological etiology. Anderson and colleagues have suggested that early premenopausal exposures and developmental events have a greater impact on risk of ER− than ER+ cancer [38, 39], equally consistent with our findings.

Combined Grade and ER expression data were not available for half the cases, although at least one was known for 84–89%. This reflects historic registry policies more than lack of diagnostic completeness, and missingness was almost equal by race within state. Facilities underreporting these prognostic details (such as non-ACoS facilities) tend to be smaller and probably serve more rural communities, so rural cases may have been subject to more imputation. Diagnosis may have been somewhat more timely at ACoS facilities (SC), but this is not suggestive of bias differential on race. We were not able to evaluate this in Ohio and do acknowledge it may have influenced our comparisons of rate differences between these two states. However, because our imputation method incorporated stage and given the results of our sensitivity analysis, we are comfortable that the potential impact of any methodologic systematic errors is small. Nonetheless, modest differences between populations, especially from about age 75 and in ER−/G1,2 cancers where numbers were small, should be viewed with caution.

Ohio does not have a reciprocity agreement to allow inclusion of cases in residents diagnosed and treated elsewhere. Any resulting underestimation of incidence rates due to more affluent patients selecting other states for both breast cancer diagnosis and treatment are likely minimal, given the excellence of the Ohio medical and cancer facilities. If any such deficits exist, they would likely result in modest underestimation of ER+ incidence especially in EA women, given its increased risk with higher socioeconomic status [40, 41].

The greater AA incidence of ER+/G1,2 cancers in Ohio suggests that as AA women become more urbanized or adopt reproductive patterns more typical of EAs, an increased incidence of ER+/G1,2 disease approximating EA rates may be seen. The currently lower AA rates, particularly in SC, may reflect a temporal delay in social changes associated with increases in ER+ cancer observed in EA women over the past 20 years [36], including a lower historical usage of hormone replacement therapy. This observation needs to be examined using data from other parts of the US, because of implications for future racial breast cancer disparities: if increases in AA incidence of ER+/G1,2 cancer are not substantially offset by reduced incidence of ER−/G3 cancer (and ER+/G3), total breast cancer incidence in AAs may increase toward that in EAs.

We suggest that the evaluation of trends in breast cancer racial disparities would be improved by examining age-adjusted and age-specific incidence rates of each ER/Grade subtype whenever possible. Routinely including combined hormone receptor expression and histologic grade (and HER-2/neu status) in association studies should also help to clarify etiologic factors for specific subtypes of breast cancer and their impacts on prognosis. Estimation of subtype incidence rates and patterns using imputed data from non-SEER registries may facilitate additional useful comparisons over time and place.

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References

1. Henson DE, Chu K, Levine PH (2003) Histologic grade, stage, and survival in breast carcinoma. Comparison of African American and Caucasian women. Cancer 98:908–917
2. Curtis E, Quale C, Haggstrom D, Smith-Bindman R (2008) Racial and ethnic differences in breast cancer survival: how much is explained by screening, tumor severity, biology, treatment, comorbidities, and demographics? Cancer 112:171–180
3. American Cancer Society (2007) Breast cancer facts & figures 2007–2008. American Cancer Society, Inc, Atlanta p. 3
4. Fitzgibbons PL, Page DL, Weaver D, Thor AD, Allred DC, Clark GM, Ruby SG et al (2000) Prognostic factors in breast cancer. College of American Pathologists Consensus Statement 1999. Arch Pathol Lab Med 124:966–978
5. Greene F, Page DL, Fleming ID, Fritz A, Balch DM, Haller DG, Morrow M (eds) (2002) AJCC cancer staging manual. Springer-Verlag, New York
6. Volpi A, Bacci F, Paradiso A, Saragoni L, Scarpi E, Ricci M et al (2004) Prognostic relevance of histologic grade and its components in node-negative breast cancer patients. Mod Pathol 17:1038–1044
7. Pinder SE, Murray S, Ellis IO, Trihia H, Elston CW, Gelber RD et al (1998) The importance of the histologic grade of invasive breast carcinoma and response to chemotherapy. Cancer 83:1529–1539
8. Page DL, Gray R, Allred DC, Dressler LG, Hatfield AK, Martino S et al (2001) Prediction of node-negative breast cancer outcome by histologic grading and S-phase analysis by flow cytometry: an Eastern Cooperative Oncology Group Study. Am J Clin Oncol 24:10–18
9. Sohib LH, Wittekind C (2002) TNM classification of malignant tumours: International Union Against Cancer
10. Cunningham JE, Butler WM (2004) Racial disparities in female breast cancer in South Carolina: clinical evidence for a biological basis even in small tumors. Breast Cancer Res Treat 88:161–176
11. Carey LA, Perou C, Livasy CA, Dressler LG, Cowan D, Conway K et al (2006) Race, breast cancer subtypes, and survival in the Carolina Breast Cancer Study. JAMA 295:2492–2502

12. Bauer KR, Brown M, Cress RD et al (2007) Descriptive analysis of estrogen receptor (ER)-negative, progesterone receptor (PR)-negative, and HER2-negative invasive breast cancer, the so-called triple-negative phenotype: a population based study from the California cancer Registry. Cancer 109:1721–1728
13. United States Department of Health and Human Services (US DHHS), Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS) (2008) Bridged-race population estimates, United States July 1st resident population by state, county, age, sex, bridged-race, and Hispanic origin, on CDC WONDER On-line Database. Vintage 2006: years 2000–2006, September 2007 online database, based on the August 16, 2007 data release. http://wonder.cdc.gov/bridged-race-v2006.html
14. National Center for Health Statistics (NCHS) (2008) http://www.cdc.gov/nchs/about/major/dvs/popbridge/popbridge.htm
15. National Cancer Institute Surveillance Research Program (2008) http://seer.cancer.gov/stdpopulations
16. North American Association of Central Cancer Registries (2008) http://www.cancer-rates.info/naaccr/
17. Jensen OM, Parkin DM, MacLennan R, Muir CS, Skeet RG (eds) (1991) Cancer registration: principles and methods. Scientific Publication No. 95. International Agency for Research on Cancer. Lyon, France, p 138 (Equations 11.11 and 11.12)
18. Sørlie T, Perou C, Tibshirani R, Aas T, Geisler S, Johnsen H et al (2001) Gene expression patterns of breast carcinomas distinguish tumor subclasses with clinical implications. Proc Natl Acad Sci USA 98:10869–10874
19. Sørlie T, Wang Y, Xiao C, Johnsen H, Naume B, Samaha RR, Børresen-Dale AL (2006) Distinct molecular mechanisms underlying clinically relevant subtypes of breast cancer: gene expression analyses across three different platforms. BMC Genomics 7:127
20. Reis-Filho JS, Tutt A (2008) Triple negative tumours: a critical review. Histopathology 52:108–118
21. Rakha EA, Ellis IO (2009) Triple-negative/basal-like breast cancer: review. Pathology 41(1):40–47
22. Asselin-Labat ML, Shackleton M, Stingl J, Vaillant F, Forrest NC, Eaves CJ et al (2006) Steroid hormone receptor status of mouse mammary stem cells. J Natl Cancer Inst 98:1011–1014
23. Sims AH, Howell A, Howell SJ, Clarke RB (2007) Origins of breast cancer subtypes and therapeutic implications. Nat Clin Pract Oncol 4:516–525
24. Melchor L, Benitz J (2008) An integrative hypothesis about the origin and development of sporadic and familial breast cancer subtypes. Carcinogenesis 29(8):1475–1482
25. Bloushtain-Qimron N, Yao J, Shipitsin M, Maruyama R, Polyak K (2009) Epigenetic patterns of embryonic and adult stem cells. Cell Cycle 8(6):806–817
26. Abdel-Fatah TMA, Powell D, Hodi Z, Reis-Filho JS, Lee AHS, Ellis IO (2008) Morphologic and molecular evolutionary pathways of low nuclear grade invasive breast cancers and their putative precursor lesions: further evidence to support the concept of low nuclear grade breast neoplasia family. Am J Surg Pathol 32(4):513–523
27. Tlsty TD, Coussens LM (2006) Tumor stroma and regulation of cancer development. Ann Rev Pathol 1:119–150
28. Hunter KW, Crawford NP (2006) Germline polymorphism in metastatic progression. Cancer Res 66:1251–1254
29. Crawford NPS, Hunter K (2006) New perspectives on hereditary influences in metastatic progression. Trends Genet 22:555–561
30. Potter J, Cerhan JR, Sellers TA, McGovern PG, Drinkard C, Kushi LR, Folsom AR (1995) Progesterone and estrogen receptors and mammary neoplasia in the Iowa Women's Health Study: how many types of breast cancer are there? Cancer Epidemiol Biomark Prev 4:319–326
31. Li CI, Malone K, Porter PL, Weiss NS, Tang M-TC, Daling JR (2003) The relationship between alcohol use and risk of breast cancer by histology and hormone receptor status among women 65–79 years of age. Cancer Epidemiol Biomark Prev 12:1061–1066
32. Smigal C, Jemal A, Ward E, Cokkinides V, Smith R, Howe HL, Thun M (2006) Trends in breast cancer by race and ethnicity: update 2006. CA Cancer J Clin 56:168–183
33. Chen WY, Colditz G (2007) Risk factors and hormone-receptor status: epidemiology, risk-prediction models and treatment implications for breast cancer. Nat Clin Pract Oncol 4:415–423
34. Yang XR, Sherman ME, Rimm DL, Lissowska J, Brinton LA, Peplonska B, Hewitt SM, Anderson WF, Szeszenia-Dabrowska N, Bardin-Mikolajczak A, Zatonski W, Cartun R, Mandich D, Rymkiewicz G, Ligaj M, Lukaszek S, Kordek R, Garcia-Closas M (2007) Differences in risk factors for breast cancer molecular subtypes in a population-based study. Cancer Epidemiol Biomarkers Prev 16(3):439–443
35. Setiawan VW, Monroe KR, Wilkens LR, Kolonel LN, Pike MC, Henderson BE (2009) Breast cancer risk factors defined by estrogen and progesterone receptor status: the multiethnic cohort study. AJE Adv Access
36. Stead L, Lash TL, Sobieraj J, Chi D, Westrup J, Charlot M, Blanchard R, Lee JC, King T, Rosenberg CL (2009) Triple negative breast cancers are increased in black women regardless of age or body mass index. Breast Cancer Res 11(2):R18 [Epub ahead of print.]
37. Karp SE, Tonin PN, Begin LR, Martinez JJ, Zhang JC, Pollak MN, Foulkes WD (1997) Influence of BRCA1 mutations on nuclear grade and estrogen receptor status of breast carcinoma in Ashkenazi Jewish women. Cancer 80:435–441
38. Anderson WF, Chen B, Brinton LA, Devesa SS (2007) Qualitative age interactions (or effect modification) suggest different cancer pathways for early-onset and late-onset breast cancers. Cancer Causes Control 18:1187–1198
39. Anderson WF, Matsuno R (2006) Breast cancer heterogeneity: a mixture of at least two main types? J Natl Cancer Inst 98:948–951
40. Krieger N, Chen JT, Ware JH, Kaddour A (2008) Race/ethnicity and breast cancer estrogen receptor status: impact of class, missing data and modeling assumptions. Cancer Causes Control doi: 10.1007/s10552-008-9202-1
41. Robert SA, Strombom I, Trentham-Dietz A et al (2004) Socioeconomic risk factors for breast cancer: distinguishing individual- and community-level effects. Epidemiology 15:442–450